

Semivra Libellus POS

Update & Changes Report

Version 2.0 Release Notes | May 29–30, 2026

System	Semivra Libellus POS
Client / Brand	Kasa Lokal
Report Type	Update & Changes Report
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Stack	React 18 + Vite + Tailwind / Express + MongoDB + Socket.io

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SECTION 1: EXECUTIVE SUMMARY

This report documents the Version 2.0 release of the Semivra Libellus POS system for Kasa Lokal. The update represents a major enterprise-readiness milestone, covering full-stack security hardening, complete accounting infrastructure, advanced inventory management with FEFO multi-batch expiry tracking, display-unit normalization (kg/L/pcs), Excel stock-take import, delivery and dispatch pipeline for Grab/Foodpanda/Manual orders, shift management UI upgrades, Docker containerization with CI/CD, operations runbook and automation scripts, structured JSON logging, and comprehensive UI/UX polish including WCAG accessibility fixes. The test suite grew from 16 to 75 passing tests, and all three build targets (server node --check, vitest, vite build) are verified clean.

Metric	Value
Tests passing	75 / 75
New API endpoints added	14+
Security vulnerabilities fixed	12+
AdminDashboard size	~4,825 lines
Deployment method	Docker + docker-compose + GitHub Actions CI
Build status	server node --check OK vitest 75/75 vite build clean

SECTION 2: NEW FEATURES

2.1 Delivery & Dispatch Pipeline

- POS checkout collects delivery details (address, phone, fee, scheduled time) for Manual Delivery and Pickup orders
- Order fulfillment modes: Dine-In, Takeout, Pickup, Manual Delivery, Grab Delivery, Foodpanda
- Dispatch status tracker on order cards: Preparing → Out for Delivery → Awaiting Pickup → Delivered/Picked Up
- OrderSchema new fields: deliveryAddress, customerPhone, deliveryFee, scheduledTime, dispatchStatus
- Receipts updated to show delivery address, phone, fee, and scheduled time

2.2 Inventory: Multi-Batch FEFO Expiry Tracking

The inventory system now supports multi-batch tracking with First-Expired-First-Out (FEFO) consumption logic, ensuring the oldest stock is always consumed first during order completion and spoilage.

- InventorySchema: added expiryBatches array (qty, expiryDate, receivedAt, reference, unitCost)
- expiryDate field now derived as soonest expiry across all batches
- server/lib/expiry.js — pure FEFO helpers: soonestExpiry, sortBatchesFEFO, consumeBatches, addBatch, batchesTotal
- 17 new vitest tests on batch math (FEFO consumption, soonest detection, integration scenario); total: 75 passing
- Order completion (ERP engine): FEFO-consumes batches after atomic stockQty deduction
- Spoilage handler: FEFO-consumes batches (oldest first — natural waste pattern)
- Procurement: seeds initial batch when expiry provided; Restock: APPENDS new batch (no longer overrides)
- New endpoints (superadmin): POST /api/inventory/:id/batches, DELETE /api/inventory/:id/batches/:batchIdx
- Frontend: inventory table row expands to show all batches sorted FEFO with "NEXT" badge on oldest
- Excel import: APPENDS new batch when row has expiry + positive diff; FEFO-consumes on negative diff

2.3 Inventory: Display Units (kg/L/pcs Everywhere)

INFO: FORCED RULE: Only kg, L, and pcs are displayed to operators across the entire app. g and ml are never shown in the UI. Storage remains in base units (g/ml/pcs) so recipes keep full precision.

- InventorySchema: added displayUnit + unitMultiplier (base units per displayUnit)
- server/lib/units.js — UNIT_TABLE, resolveUnit, displayToBase, baseToDisplayCost helpers
- 19 new vitest tests for unit conversion (canonical, case-insensitive, round-trip)
- effectiveDisplay(item) helper auto-promotes g→kg, ml→L when displayUnit is missing
- Procurement form and Edit modal unit dropdowns: g/ml removed (kg/L/pcs only)
- Inventory table shows qty, unit cost, and low-stock threshold all in display unit
- SRP field removed from Edit modal and Excel import template (not needed for inventory management)
- Excel template downloads with new 5-column header: Code, Product, Qty Unit, Unit Cost, Expiry date
- Server import endpoint auto-promotes incoming g→kg / ml→L

2.4 Inventory: Excel Stock-Take Import

- New endpoint POST /api/inventory/import (superadmin, transactional, max 2,000 rows)
- Excel standard header: Code, Product, Qty Unit, Unit Cost, Expiry date

- Product column parses trailing pack-size hint: "Milk 1L" → itemName="Milk", unit hint=L
- Existing items: stockQty replaced with new value; diff booked as journal entry
 - Positive diff: DR 1500 Inventory Asset / CR 4200 Inventory Adjustment Gain
 - Negative diff: DR 5100 Spoilage, Variance & Waste Expense / CR 1500 Inventory Asset
- New items: auto-created with itemCode; booked DR 1500 / CR 4200
- Every row writes a StockCard entry (type: "Stock Take Import")
- Frontend: Import button (file picker .xlsx/.xls/.csv) + Template button (sample CSV download)
- Client-side parse via SheetJS, lazy-loaded only when importing
- Preview modal with color-coded diff table: NEW / ↑ INC / ↓ DEC / SAME / ERROR rows
- Confirmation dialog before posting; results modal shows totals

2.5 Inventory: Expiry Monitoring

- InventorySchema: added expiryDate (Date, indexed) + expiryWarnDays (default 7)
- New endpoint GET /api/inventory/expiring?days=30 — returns expired / expiringSoon / expiringLater
- New endpoint PATCH /api/inventory/:id/expiry — update expiry / warn-days only
- Inventory table: Expiry column with color-coded badges — EXPIRED (red pulse), TODAY, ≤warn days (yellow), ≤30d (orange), >30d (gray)
- Expiry Watch panel in inventory right column — lists items expiring within 30 days, sorted by days remaining
- Sidebar Inventory badge combines low-stock + expired + expiring-soon counts; pulse animation when any expired item exists
- Waste modal: smart-defaults to Spoilage reason + full stockQty + auto-flag note when item is already expired
- Spoilage flow: auto-clears expiryDate when item.stockQty hits 0 (next restock sets a fresh date)

2.6 Waste / Spoilage Logging

- POST /api/inventory/spoilage/:id endpoint + dedicated modal in Inventory tab
- Enforced reason dropdown in spoilage form (Expired, Damaged, Overproduction, etc.)
- Spoilage auto-creates StockCard entry (type: "Spoilage")
- Journal entry: DR 6200 Spoilage Expense / CR 1500 Inventory Asset
- FEFO batch consumption: oldest batches depleted first during spoilage

2.7 Shift Management Upgrade

- GET /api/shifts endpoint (superadmin only, paginated) — full shift history archive
- Shift History sub-tab added under Daily History & Shifts tab — shows full cashier shift ledger with variance colors
- X-Reading button on History tab — generates PDF mid-shift summary without closing the register

2.8 Accounting: Full Chart of Accounts + Financial Reports

A complete double-entry accounting infrastructure has been implemented, from canonical chart of accounts through to P&L; and Balance Sheet aggregation pipelines.

Account Range	Type	Examples
1xxx	Assets	1000 Cash on Hand, 1200 AR, 1500 Inventory
2xxx	Liabilities	2000 AP, 2100 (retired - VAT)
3xxx	Equity	3000 Owner Equity, 3100 Retained Earnings
4xxx	Revenue	4000 Sales Revenue (Non-VAT)
5xxx	COGS	5000 COGS, 5100 Spoilage & Waste
6xxx	Operating Expenses	6000 Salaries, 6100 Complimentary, 6200 Spoilage

- `server/lib/chartOfAccounts.js` — canonical chart with 17 vitest tests (account-type coverage, code-range classifiers, mutual exclusivity)
- New endpoints (all superadmin): `GET /api/expenses/categories`, `POST /api/expenses`, `POST /api/orders/:id/settle-ar`, `GET /api/finance/ar-outstanding`
- New report endpoints: `GET /api/reports/pnl?start=&end;=` (MongoDB aggregation P&L;), `GET /api/reports/balance-sheet?asOf=` (aggregation Balance Sheet)
- `GET /api/journal/export?start=&end;=` — streams CSV download of journal entries
- Frontend Ledger tab: 5 sub-tabs (Journal, P&L;, Balance Sheet, A/R Outstanding, Add Expense)
- P&L; view: date-range picker, revenue/COGS/OpEx breakdown, gross/net income KPI cards, CSV export
- Balance Sheet: three-column Asset/Liability/Equity tables + balanced indicator
- A/R Outstanding view: per-order Settle action with modal (amount, deposit method, note)
- Expense entry modal: amount, category, paid-from, description, vendor, date

2.9 E-Wallet + Bank — Accounts Receivable Policy

INFO: POLICY CHANGE: Only physical Cash (payment method = "Cash") hits Account 1000 Cash on Hand immediately. All other payment channels (Bank Transfer, GCash, Maya, Maribank, E-Wallet, Grab Delivery, Foodpanda, Manual Delivery) now book to Account 1200 Accounts Receivable until the merchant manually verifies receipt and settles via the A/R settlement modal.

- `server/lib/ledger.js:debitAccountFor(method)` simplified — Cash → 1000, all else → 1200
- `suggestedSettleAccount(method)` helper provides smart UI defaults for settlement destination
- `SETTLE_DESTINATIONS` array covers accounts 1000/1010/1015 with match arrays
- Void handler: new guard refuses to void already-settled A/R orders (would orphan paired journal entries)
- 6 new vitest tests — Bank Transfer/E-Wallet/Delivery all assert → 1200; settle account mappings tested
- Frontend A/R tab heading updated to "Non-Cash sales awaiting settlement (E-Wallet · Bank · Delivery)"
- Settle button smart-defaults "Deposited To" to match original payment channel

2.10 NON-VAT Compliance Badges

- "NON-VAT REGISTERED" badge added to sidebar brand section and mobile header
- Journal entries no longer create phantom Account 2100 VAT Payable entries (business is Non-VAT registered)

- Receipts show "NON-VAT REGISTERED" prominently
- Revenue uses gross receipts formula (total + discount, no VAT separation)

2.11 Low Stock Alert Threshold

- InventorySchema: added `lowStockThreshold` field
- Alert badge on sidebar Inventory nav and inventory table rows when `stockQty ≤ threshold`
- Procurement form includes Low Stock Alert Threshold input field
- Low Stock Alerts panel added to inventory right column with item list

SECTION 3: PRODUCTION INFRASTRUCTURE

3.1 Docker & Containerization

- `server/Dockerfile` — multi-stage build; production-only dependencies
- `client/Dockerfile` — multi-stage build (Vite build stage + nginx serve stage)
- `docker-compose.yml` — orchestrates server + client with healthchecks
- `client/nginx.conf` — SPA fallback (`index.html` for unknown routes) + asset cache headers
- `.dockerignore` — excludes `node_modules`, `.env` files, logs

3.2 CI/CD (GitHub Actions)

- `.github/` — workflows for server (`node --check + vitest`) and client (`vite build`)
- Both workflows run on push and `pull_request` to main branch
- `npm audit --audit-level=high --omit=dev` added to both server + client CI jobs

3.3 Operations Scripts & Runbook

- `scripts/setup-env.sh` — interactive `.env` generator; openssl-generated `JWT_SECRET` (64 hex chars) + `ADMIN_PASS` (16 chars); prompts for `MONGO_URI` + `ALLOWED_ORIGINS`; writes `server/.env` at mode 600
- `scripts/backup-mongo.sh` — daily `mongodump` to `backups/YYYY-MM-DD_HHMMSS.tar.gz`; configurable `RETENTION_DAYS` (default 30); optional S3 upload via `BACKUP_S3_BUCKET`; cron-ready
- `scripts/restore-mongo.sh` — interactive backup picker with typed-confirm guard; `--drop` restore
- `Makefile` — `make setup`, `make deploy`, `make health`, `make backup`, `make restore`, `make logs`, `make status`, `make test`; self-documenting `make help`
- `DEPLOY.md` — Day-1 deployment runbook with pre-flight checklist, smoke test, Week-1 verification, incident response playbook, env var reference
- `server/.env.example` — all required + optional env vars documented

3.4 Structured Logging

- `pino` + `pino-http` installed; request-level JSON logging (skips `/health` endpoint to reduce noise)
- Key `console.*` calls replaced with structured log calls (`level`, `timestamp`, `requestId`)
- `uncaughtException` + `unhandledRejection` handlers added for crash-safe process lifecycle

3.5 Health Endpoint

- `GET /health` (no auth required) — returns DB `readyState` + server uptime; suitable for load balancer health checks

SECTION 4: SECURITY FIXES

4.1 Critical Privilege Escalation Fixed

CRITICAL: CRITICAL: PUT /api/users/:id was accessible by any authenticated staff member, allowing a regular cashier to change the superadmin password. requireSuperAdmin middleware has been added.

- PUT /api/users/:id: added requireSuperAdmin — any staff could previously change superadmin password
- Added requireSuperAdmin to: POST /api/journal, POST /api/users, DELETE /api/users/:id
- Added requireSuperAdmin to: POST /api/orders/:id/void (any staff could previously void completed orders)

4.2 Name-String Bypass Eliminated (All Instances)

CRITICAL: CRITICAL: Role authorization was checking name === "Super Admin" as a fallback alongside role === "superadmin". Any user who created an account with the name "Super Admin" could bypass superadmin guards. All 7 occurrences have been removed — role-only check now enforced everywhere.

- SuperAdminPanel.jsx line 40: user.name === 'Super Admin' → user.role === 'superadmin' (UserCard protection)
- SuperAdminPanel.jsx line 176: removed name fallback in auth state initializer
- SuperAdminPanel.jsx line 267: removed name fallback in selectableUsers batch protection
- SuperAdminPanel.jsx line 283: removed name fallback in login handler
- AdminDashboard.jsx fetchUsers: u.name !== 'Super Admin' → u.role !== 'superadmin'
- Server-side seeder: existing "Super Admin" users without role field get role: 'superadmin' set on every boot

4.3 Other Security Hardening

- Cashier impersonation prevented: cashier in POST /api/orders now reads from req.user?.name (JWT), not req.body
- Removed dead ADMIN_PIN = '1234' and SECRET_TOKEN = 'cafe2026' constants from AdminDashboard.jsx (were visible in browser bundle)
- GET /api/inventory/history and GET /api/inventory/history/:id now protected by verifyToken (were publicly accessible)
- QR session ID generation: Math.random() → crypto.randomBytes(32).toString('hex')
- Rate limiting: loginLimiter (10 attempts / 15 min) + orderLimiter (60 orders / min)
- Removed dead cashier field from PUT /api/orders/:id req.body destructure

SECTION 5: PERFORMANCE OPTIMIZATIONS

5.1 Backend

- Finance balances endpoint rewritten as MongoDB aggregation pipeline — no longer loads all journal entries into memory (OOM-safe at scale)
- Removed duplicate `app.use(express.json())` middleware call
- N+1 fix in order completion: bulk pre-fetch products (1–2 queries instead of N), batch `StockCard.insertMany` (1 query instead of M)
- Compound indexes added: `Order(status+isArchived)`, `Order(createdAt)`, `Product(name, category)`, `JournalEntry(date, reference)`, `User(name)`
- Inventory pagination: `GET /api/inventory` supports `?page=&limit;=&search;=`
- Journal pagination: `GET /api/journal` always paginated (default 50/page)

5.2 Frontend

- Countdown extracted into isolated `MidnightCountdown` component — eliminates 1-second full re-renders of the 4,825-line `AdminDashboard`
- Analytics engine wrapped in `useMemo([orders, archivedOrders, inventory, products])` — no longer recomputes on every timer tick
- `dailyRevenueList.reverse()` → `[...dailyRevenueList].reverse()` — fixes in-place mutation of memoized array causing alternating order on re-render

SECTION 6: BUG FIXES

Area	Fix
Analytics — Gross inflation	<code>todayGross</code> now excludes complimentary orders (was incorrectly inflating the gross figure)
Analytics — Revenue formula	<code>todayRevenue = todayGross - todayDiscounts</code> (was double-subtracting the complimentary amount)
Non-superadmin force logout	<code>fetchOrders</code> now guards archives call with <code>activeAdmin?.role === "superadmin"</code> check — previously non-superadmin users were instantly force-logged-out on every order fetch because <code>GET /api/orders/archives</code> requires <code>requireSuperAdmin</code> and <code>apiFetch</code> logs out on 403
Socket duplicate listeners	Socket bug fixed: both <code>useEffects</code> registered <code>"orderUpdated"/"erpUpdated"</code> ; <code>socket.off(event)</code> in cleanup removed ALL listeners. Now uses named callbacks + <code>socket.off(event, fn)</code> for scoped removal
Orders pagination filter	Orders pagination now uses <code>displayOrders</code> (filtered array) not the unfiltered orders array; pagination was showing wrong page when filters were active
Void add-on inventory	Void handler: add-on ingredient inventory restoration loop was missing — voids now correctly restore add-on stock + write <code>StockCard</code> entries

Complimentary journal	Complimentary order journal: removed phantom DR 6100 / CR 4000 entries at selling price; now only records DR 6100 / CR 1500 at cost
EOD display units	EOD audit: Physical Count input accepts display values (kg/L) and is converted to base units on submit
EOD footer label	"Net Variance" relabelled to "Items With Variance" (counts items with variance, not raw quantity)
handleBankDeposit	Added null guard on shiftReconcile.result to prevent crash when reconciliation modal closes early
handleSaveProduct	Now checks response.ok and alerts the operator on server error (was silently failing)
fetchEODData	Added res.ok guard before res.json() to prevent JSON parse error on 4xx/5xx responses
QR table ID	handleShowQR: Math.random() → Date.now().toString(36) to avoid collisions on rapid scans
Dead body fields	Removed dead cashier field from submitManualOrder and updateStatus payloads (server reads from JWT)

SECTION 7: UI / UX CHANGES

7.1 POS Register Rebuild

- Real-time product search bar on POS register — instant filter as operator types
- Enlarged touch targets for tablet hardware: minimum 44px hit area on all interactive elements
- Inline discount row (■ flat or % percentage) directly on POS checkout

7.2 Payment Checkout Modal

- Thermal receipt preview rendered inside the checkout modal before confirming
- Payment method pills: Cash, GCash, Maya, Maribank, Bank Transfer, E-Wallet, Grab Delivery, Foodpanda, Manual Delivery
- Quick cash denomination buttons: ■20, ■50, ■100, ■200, ■500, ■1000
- Exact / Round shortcuts for fast cash entry
- Live change calculation as cashier inputs tendered amount

7.3 Navigation Improvements

Old Name	New Name
Orders	Orders & POS
Inventory	Inventory & Stock
History	Daily History & Shifts
Ledger	Accounting & Ledger
Products	Pricing Control

7.4 Color / Theme System Fixes

- Tailwind dark token remapped to --page-bg (#0d0d0d) — was incorrectly mapped to white
- Fixed 117 occurrences of bg-dark / text-dark in AdminDashboard — were rendering white-on-white invisible text
- Added semantic foreground tokens: --fg (body text: 245 245 245) and --fg-muted (secondary: 180 180 180)
- bg-receipt / text-receipt dedicated tokens for thermal receipt printing (always white bg + black text regardless of theme)
- Fixed WCAG contrast violations on analytics cards (bg-accent + text-black → bg-brand-dark + text-white/70)
- Responsive grid fix: analytics velocity grid-cols-4 → grid-cols-2 sm:grid-cols-4 (fixes 8px text at 360px viewport)
- Added text-fg / text-fg-muted Tailwind utilities for forward use

7.5 Accessibility (ARIA)

- aria-label added to all login form inputs (AdminDashboard + SuperAdminPanel)
- aria-expanded + dynamic aria-label added to mobile menu toggle button
- aria-pressed added to CustomerMenu category filter buttons
- aria-label added to CustomerMenu cart remove buttons and customer name input

SECTION 8: REMOVED / DEPRECATED

Item	Reason
BIR 2551Q Report	Removed entirely: BIRReport.jsx deleted, /api/reports/bir endpoint removed (~158 lines), sidebar nav link removed
server/middleware/auth.js	Deleted — was dead code; inline middleware in server.js is used instead
Account 6200 Spoilage Expense	Retired — all spoilage entries now use Account 5100 "Spoilage, Variance & Waste Expense"
Account 2100 VAT Payable	Removed from ERP engine — business is Non-VAT registered; phantom VAT entries were creating incorrect journal entries
Account 6101	Retired — complimentary completion and its void reversal both now use Account 6100
SRP in Excel import	SRP (Suggested Retail Price) column removed from Excel import template — not needed for inventory stock-take
ADMIN_PIN / SECRET_TOKEN	Removed dead constants from AdminDashboard.jsx bundle (were never used, visible to anyone who inspected the JS bundle)
VAT Payable journal entries	Phantom DR/CR entries for Account 2100 removed from both order completion ERP and void handler

SECTION 9: TEST SUITE

NOTE: Build status: server node --check OK | vitest 75/75 passing | client vite build clean

Test File	Coverage Area	Tests
server/lib/ledger.js	Account mapping (debitAccountFor), balance assertion, Non-VAT gross-sales math	16
server/lib/chartOfAccounts.js	Account-type coverage, code-range classifiers, mutual exclusivity, getAccount fallback, EXPENSE_CATEGORIES integrity, Non-VAT compliance	17
server/lib/units.js	Canonical unit names, case-insensitive aliases, round-trip display-to-base conversions	19
server/lib/expiry.js	FEFO consumption order, soonest expiry detection, integration scenario (multi-batch)	17
server/lib/ledger.js (AR policy)	Bank Transfer/E-Wallet/Delivery assert → 1200; suggestedSettleAccount mappings; SETTLE_DESTINATIONS integrity	6
TOTAL		75

SECTION 10: DEPLOYMENT CHECKLIST

Follow these steps for the first production deployment. Complete each step in order.

#	Step	Action
1	Generate environment secrets	Run <code>scripts/setup-env.sh</code> — creates <code>server/.env</code> with a 64-hex <code>JWT_SECRET</code> and 16-char <code>ADMIN_PASS</code> generated by <code>openssl</code> . Never commit this file.
2	Configure MongoDB	Verify <code>MONGO_URI</code> in <code>server/.env</code> points to your production MongoDB Atlas cluster. Ensure IP whitelist includes your server's IP.
3	Set allowed origins	Set <code>ALLOWED_ORIGINS</code> in <code>server/.env</code> to your Vercel frontend domain (e.g. <code>https://your-app.vercel.app</code>). Omit trailing slash.
4	Deploy containers	Run <code>make deploy</code> (equivalent to <code>docker-compose up -d --build</code>). Both server and client containers will build and start.
5	Confirm health	Run <code>make health</code> — verify server returns <code>{"status":"ok","db":"connected"}</code> . If DB shows "disconnected", check <code>MONGO_URI</code> + Atlas whitelist.
6	Verify superadmin login	Login as Super Admin, verify the shift start modal appears and Starting Cash input works.
7	End-to-end order test	Create a test order, mark it as Ready, then Delivered/Completed. Verify journal entries appear in Accounting & Ledger tab.
8	Week-1 monitoring	Run <code>make logs</code> daily to watch for errors. Run <code>make backup</code> to confirm <code>mongodump</code> succeeds. Check sidebar expiry badge for any expired items.

INCIDENT RESPONSE QUICK REFERENCE

- Server will not start: Check `server/.env` exists and `MONGO_URI` / `JWT_SECRET` are set. Run `make logs` for stack trace.
- DB disconnected: Check MongoDB Atlas IP whitelist. Check `MONGO_URI` format (must start with `mongodb+srv://`).
- White screen on client: Open browser console. If 401 — token expired, clear `localStorage` and log in again. If 500 — check `make logs`.
- Journal imbalance error: A `balanced-journal` assertion fired. Check the order that triggered it and verify the chart of accounts is correct.
- Inventory negative: A race condition may have slipped through. Use the Restock or Spoilage modal to correct, then check recent StockCard history.